Classical cuts: source evidence and information at an interface

The source-supported common question is how contacts, sections or local observations compose into a whole. It supplies useful modern model questions, not a theorem requiring a positive action, quantum dynamics or Newtonian anticipation of them. Aristotle's possible-versus-simultaneous division argument, Plutarch's cone report and the contact arguments below have distinct premises. Similarity documents no transmission.

The held editions and source companions determine the reading coverage. Passage interpretations are distinguished below from incomplete witnesses and retrieval leads. The independent mathematical result about bounded-speed scalar kernels is maintained in finite propagation.

Possible divisions, cone sections and the arrow

Aristotle, *Generation and Corruption* I.2, around 317a, challenges the passage from division at any chosen point to simultaneous division everywhere, using the absence of immediately consecutive points; earlier he treats contact as a relation of two things. The Joachim passage companion and French translation, paragraphs 10–21 are the anchors. Paragraph 19 presents an atomist inference examined by the following text, not an isolated statement of Aristotle's final view. The editors acknowledge interpretive difficulty. This is the principal source for the anywhere-versus-all-together comparison.

Plutarch, *De communibus notitiis* 39, 1079e–1080a, reports Democritus's cone dilemma and Chrysippus's reply in a hostile discussion. Equal sections suggest a cylinder; unequal ones suggest steps. The Goodwin/Perseus section retains the edition anchors. The Xylander witness locates it in the 1570 *Moralia*, pp. 823–824. That source search did not verify a particular early-modern mathematician using this passage for indivisibles; the separate rolling-cone/Leibniz comparison establishes no such reception.

A modern continuum distinction separates a common boundary, sections at different heights and their limiting variation. For $A(z) = \pi r^2(1 - z/h)^2$, $A(z + \delta) - A(z) = A'(z)\delta + O(\delta^2)$; the difference vanishes while its ratio to δ need not. There is no next distinct real height. This calculation is a modern comparison, not a translation of Chrysippus into differential geometry.

Rivero's 1999 cone essay is a modern source for keeping finite separation distinct from a rescaled difference; it is not an ancient witness or evidence of Newtonian reception.

Aristotle's *Physics* VI.9 diagnoses the arrow inference as taking time to consist of nows. The Teubner companion records damaged Greek OCR and limits the reading. Diogenes Laertius IX.72 reports the locative dilemma under Zeno's name; related Diodorus attribution is a separate question. A position specifies occupancy; velocity, momentum and a transition rule relate neighboring states. An ordinary classical phase state already supplies this information. The proposal that the cone and arrow are related by "Wick rotation" is a modern heuristic. Analytic continuation of chosen telegraph and Dirac models establishes no identity between historical arguments or physical stochastic and quantum theories.

Contacts and spatial composition across traditions

Vasubandhu's *Vimsatika* 12 asks how a partless atom joins six neighbors: distinct directional contacts give it parts, while putting all six in one location leaves an aggregate only atom-sized. Verse 13 tests the reply that aggregates join although atoms do not. Its setting is an argument about external perceptual objects in vijñaptimātra. Ibn Sina's *Physics* III.4, in the McGinnis translation indexed at pp. 75–76, examines differentiated contact and interpenetration; the middle member of a three-body chain must distinguish its two contacts. The Indian and Avicennian companions preserve the passages and their different access levels. The structural parallel supplies no transmission route.

The modern model distinction is among a partless constituent, distinguishable contact locations and greater extension of the composite. Ambient geometry or explicit interface variables can carry the missing relations. A physical argument must specify which resources it permits before claiming that classical composition is impossible.

Liu Hui's received commentary in *Nine Chapters*, juan 5, explains a 2:1 volume ratio by dissection and successively finer remainders. The solid companion contains an unproofread facsimile transcription, not a completed scan audit. Its pertinent question is control of the total remainder, rather than a generic resemblance to infinitesimals.

Mohist B17 distinguishes a re-created shadow from a transported object; B25a/b examine rigidity and the changed effectiveness of a weight as the apparatus configuration changes. The locally held Schemmel–Boltz 2023 edition and selected pages separate the ancient witness from modern restorations. These are useful body/effect and body/apparatus distinctions. The Zhuangzi 33 witness places the flying shadow, arrow and endlessly halved stick among the debaters' theses; do not assign every listed thesis to Hui Shi personally.

Nagarjuna's *Mulamadhyamakakarika* II.1–3 tests traversal in the traversed, untraversed and being-traversed. II.3 exposes that being-traversed already presupposes traversal (Indian passage companion). The modern question is what whole-path relations a local predicate has already borrowed. It is distinct from the atom-contact argument.

Cohesion, time atoms and limiting ratios

Galileo's *Two New Sciences*, First Day, Crew–De Salvio 1914, pp. 18–19, proposes small vacua as a speculative account of cohesion. The OCR and companion also locate the wheel and bowl/cone passages. The relevant question is what interaction remains at an apparently empty interface. This vacuum is not the modern quantum vacuum.

The al-Shahrastani report on al-Nazzam combines denial of indivisible parts with the leap; the reporter objects that a leap still traverses a distance. In Maimonides I.73, time atoms, unequal speeds and the rotating millstone test whether a microscopic rule preserves bodily coherence. These are competing responses, not a common regional doctrine. A specifically Ibn Sinan millstone argument and a cuneiform stepped-solid volume problem remain retrieval leads; neither the remembered example nor the ziggurat image is held evidence of the desired argument.

Newton's Section I closing Scholium addresses ultimate ratios and limiting velocity. The source geometry separates the actual words from modern action reconstructions. A local polygon error may vanish while a specified ratio survives. Neither this nor the contact analogies establishes that Newton suspected a universal action scale or a fluxional-time quantum interpretation.

Refinement, elimination and physical records

Inserting y between x, z replaces a segment by two segments and changes the velocity assignments as y varies. Three operations must be distinguished:

- Sampling a fixed trajectory leaves the old positions unchanged.
- Re-solving an approximate boundary-value or variational problem on a finer mesh can change its old-time vertices. This compares mathematical solutions; it does not change the physical past.
- Inserting a variable into an exact consistent path law preserves the old marginal after integration. Conditioning on its observed value can change predictions for correlated variables; physical intervention changes the law.

The one-cut action calculation fixes both endpoints and does not prove readjustment of every free vertex. Composing full classical segments before eliminating interfaces can differ

from eliminating each interface first and composing only the reduced states. Ordinary hidden classical memory can explain this difference; an action floor requires an independently justified restriction on that memory or its records. The receiver and readout constructions retain explicit apparatus, vanishing-disturbance preparations and the difference between actual new probes and refinement of reported samples. They remain countermodels to an automatic lower bound from adding cuts.

Fixed-time compatibility and calibration

A separate proposed question asks whether physically motivated compatibility rules for measurement arrangements force more than a classical joint state. Specify the allowed preparations, responses, overlap restrictions and repeatability, including apparatus variables. Agreement on overlaps need not by itself supply a global extension; assuming that an outcome is independent of arrangement is an extra premise, not a consequence of classical mechanics. A comparison must identify the rule that excludes the explicit classical model. The original Kochen–Specker construction remains a primary-source lead here; no contextuality theorem has been verified or applied to these mechanics models. A dimensionless compatibility obstruction would still need a mechanical calibration to select an action. Positivity, universality and identification with physical $\hbar$ are three further obligations.

Modern discrete-state and operator comparisons

The earlier conversation's list of discrete-substrate authors was unverified recall, not source evidence for a scientific conclusion. Its useful question is whether loss of classical state information supplies an action restriction. The relevant named source leads are 't Hooft's 1988 equivalence relations, 1999 dissipative system and 2016 book (metadata only). No universal time-grain or information-loss necessity is established from those unread arguments. A finite reversible permutation already has a unitary realization with a self-adjoint logarithm whose spectrum can be bounded below; the phase eigenvalues leave logarithm branches and physical energy normalization undetermined. The former blanket assertion that a reversible automaton has no ground state was too strong.

A discrete tick and a force-dependent record resolution make different claims. The conditional mark bounds neither require a universal time atom nor rule out an independently supplied smaller one. The thermodynamic inequality prices a stipulated information/work trade-off without deriving an absolute floor. The superdeterminism model has its own setting-device mixing and resolution assumptions; it is not a theorem about every deterministic substrate.

Dedekind's cut specifies a real through two rational sets; a finite recorded interval is a different object. The cut-energy proof counts cuts only after a positive record-share floor is supplied. Similarly, writing a spectral-triple differential as $[D, f]$ does not remove points: commutative function algebras still have evaluation states. A distance formula using $\|[D, f]\| \leq 1$ requires its spectral-triple assumptions. Setting $p = \hbar D$ supplies an action conversion rather than deriving one. These are modern mathematical comparisons, not ancient transmission claims.

The groupoid state theorem separates geometric $\epsilon = \hbar/S_0$ from duration and mesh. A mark decision window cannot be declared a distance from its classical fiber by identifying all three parameters. The normalized free blocking test admits both zero and positive action coefficients; it supplies a concrete countertest to automatic scale selection.

Consequence for STATE

The historical witnesses specify questions about division, interfaces, remainder control and local descriptions. Their unique premises survive without the old idea/task instructions. Exact sampling, re-solving, marginalization and physical measurement have different composition rules.

The separate finite-propagation proof and classical apparatus countermodels set the mathematical boundaries; no historical analogy selects physical $\hbar$.